Alessio Ferrari
Gouri Ginde *Editors*

Handbook on Natural Language Processing for Requirements Engineering

Springer

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Editors

Alessio Ferrari
University College Dublin (UCD)
Dublin, Ireland

Gouri Ginde
Electrical and Software Engineering,
Schulich School of Engineering
University of Calgary
Calgary, AB, Canada

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Foreword

Natural Language Processing (NLP) has long played a crucial role in Requirements Engineering (RE), since requirements are often (partly) expressed in natural language. This book involves many of the talented experts in the RE community and provides a comprehensive, accessible treatment of the most important aspects of the application of NLP to RE. It can thus serve both as a primer to newcomers and as a reference to experts, both from academia and industry.

Academics will benefit from getting a thorough and structured treatment of the state of the art, as well as clear guidelines and recommendations to improve the way they conduct research. Practitioners will, on the other hand, get access to the most recent solutions and innovations to address their RE problems and clear practical guidelines to apply them.

The various subjects and techniques being presented in this book are core to NLP for RE. Its chapters cover important topics related to the effective engineering of requirements, such as their classification, traceability or quality assurance. They also address specific and important applications of NLP such as privacy, regulatory compliance or issue trackers. Last, they include practical guidelines for selecting RE techniques and tools, as well as performing empirical studies in a rigorous fashion. The former also tackles the rising use of Large Language Models and more generally generative Artificial Intelligence to better support RE. Taken together, these chapters constitute an all-encompassing treatment of modern RE topics.

In summary, this book is truly a tour de force I can strongly recommend to anybody interested in RE as a research subject or its effective application in practice.

August 7, 2024

Lionel Briand

IEEE Fellow
ACM Fellow
Fellow of the Academy of Science of the Royal Society of Canada

University of Ottawa, Canada, and
Lero Centre, University of Limerick, Ireland

Preface

Requirements are both the cornerstone and the pain point of software engineering. They define the functionalities and qualities that the software must exhibit to satisfy stakeholder needs, yet their incorrect analysis and management can lead to costly errors and delays in the development lifecycle. Moreover, requirements can become challenging to handle in large projects due to their tendency to change, proliferation in number, diversification in terms of abstraction levels and stakeholders' perspectives considered. Last but not least, requirements are notoriously expressed in natural language, a medium subject to multiple interpretations. This inherent ambiguity does not align well with the precision and rigour required in software development.

Requirements Engineering (RE) aims to tackle these issues systematically, incorporating tasks such as defect detection, traceability, classification and more. However, performing these tasks manually can be error-prone, as it demands meticulous attention to detail and is sometimes tedious since RE activities can often become repetitive. For these reasons, requirements engineers have long hoped for automated partners to assist them in addressing the challenges associated with natural language requirements, and relieve their daily frustrations.

Natural Language Processing (NLP) is a field of artificial intelligence focusing on developing algorithms and models to process, interpret and understand natural language data. In order to support RE tasks with automated tools and facilitate the work of requirements engineers, researchers have experimented with NLP techniques for over 20 years, tailoring existing approaches to the specific needs of the RE domain. The interest in NLP techniques for RE has recently witnessed significant growth, also driven by the advancements in NLP technologies, which have become increasingly accessible to researchers and even to the general public, with tools that perform complex tasks such as machine translation and question answering with near-human levels of performance. We do not mention ChatGPT because this name may soon become obsolete given the pace of evolution of the NLP field.

This *Handbook on Natural Language Processing for Requirements Engineering* aims to provide a comprehensive guide on how NLP can be leveraged to enhance various aspects of RE, leading the reader from the exploration of fundamental concepts and techniques to the practical implementation of NLP for RE solutions in real-world scenarios.

The motivation for this book is threefold. Firstly, the field of NLP for RE has accumulated a vast body of knowledge over the years, yet this knowledge is dispersed across various publications and venues, making it challenging for interested individuals to efficiently obtain a comprehensive overview. This book aims to consolidate this knowledge into a single, accessible source, providing readers with a clear roadmap through the complexities of integrating NLP with RE. Secondly, NLP is currently experiencing a transformative period, mainly driven by advancements in Large Language Models (LLMs). These models present unprecedented opportunities to tackle RE challenges with enhanced effectiveness. By clarifying how NLP can support RE and showcasing state-of-the-art techniques and methodologies, this book serves as a gateway for researchers and practitioners to leverage these advancements in practical applications. Lastly, throughout our career, we have witnessed firsthand the struggles practitioners face in handling natural language requirements and the potential of NLP to alleviate these challenges. We thus aim to help them understand how they can apply NLP to enhance their RE process and, ultimately, the quality of their systems.

The book is structured into three parts, each focusing on distinct aspects of applying NLP to RE:

Part I—NLP for Downstream RE Tasks This part delves into the application of NLP techniques to tasks that are typically part of the RE process. It includes chapters on NLP for requirements classification, requirements similarity and retrieval, requirements traceability, defect detection, and automated terminology and relations extraction.

Part II—NLP for Specialized Types of Requirements and Artefacts This part explores how NLP can be tailored to handle specific requirements types and artefacts. The chapters cover legal requirements processing, privacy requirements acquisition and analysis, user feedback intelligence, mining issue trackers, and analysis of user story requirements.

Part III—NLP for RE in Practice The final part addresses practical applications and tools for implementing NLP in RE. It includes a chapter on the different tools that use NLP techniques for RE tasks, followed by chapters on empirical evaluation of tools, practical guidelines for selecting and evaluating NLP techniques, guidelines on using LLMs in RE, and dealing with data challenges in RE.

This book is designed for a diverse audience. Whether you are a researcher, a practitioner, or a student in the field of RE, this volume offers valuable insights and practical guidance. The content is structured to provide a clear progression from fundamental concepts to advanced techniques, ensuring that readers of all levels can reap the benefits.

We would like to express our gratitude to all the individuals and organizations who have supported us throughout this journey. Special thanks to our colleagues, who provided valuable feedback and encouragement; to the Springer team, which has always been present to guide us throughout the publication process; and most of all, to the authors, who meticulously edited their chapters and carefully reviewed each other's manuscripts. Special thanks go to Daniel M. Berry, who helped us substantially improve the first chapter; to Quim Motger, Muhammad Abbas, and Sallam Abualhaija, who provided us with aesthetically pleasing LaTeX code that was beneficial to all the authors; and to Lionel Briand, for kindly writing a foreword to the volume.

This book reflects the outcome of thorough research, practical insights, and collaborative efforts aimed at connecting theoretical advancements in NLP with their practical applications in RE, thereby bridging the gap between theory and practice, and providing both a foundation for understanding and a catalyst for innovation. We sincerely hope that the readers will appreciate the efforts of all the people involved in making this book comprehensive and accessible.

August 7, 2024

Alessio Ferrari
Pisa, Italy

Gouri Ginde
Calgary, Canada

Table of Contents

III NLP for RE in Practice

List of Contributors

Editors

Alessio Ferrari University College Dublin (UCD), Dublin, Ireland

Consiglio Nazionale delle Ricerche (CNR), Istituto di Scienza e Tecnologie dell'Informazione "A. Faedo" (ISTI), Pisa, Italy

Gouri Ginde Electrical and Software Engineering, Schulich School of Engineering, University of Calgary, Calgary, AB, Canada

Authors

Muhammad Abbas RISE Research Institutes of Sweden, Mälardalen University, Västerås, Sweden

Sallam Abualhaija SnT, University of Luxembourg, Luxembourg, Luxembourg

Waad Alhoshan Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, Saudi Arabia

Chetan Arora Monash University, Clayton, VIC, Australia

Sarmad Bashir RISE Research Institutes of Sweden, Mälardalen University, Västerås, Sweden

Daniel M. Berry Cheriton School of Computer Science, University of Waterloo, Waterloo, ON, Canada

Volodymyr Biryuk Department of Informatics, University of Hamburg, Hamburg, Germany

Mitra Bokaei Hosseini University of Texas at San Antonio, San Antonio, TX, USA

Travis Breaux Carnegie Mellon University, Pittsburgh, PA, USA

Sjaak Brinkkemper Department of Information and Computing Sciences, Utrecht University, Utrecht, The Netherlands

Lionel Briand School of EECS, University of Ottawa, Ottawa, ON, Canada; Lero SFI centre for Software Research and University of Limerick, Limerick, Ireland

Marcello Ceci SnT, University of Luxembourg, Luxembourg, Luxembourg

Jane Cleland-Huang University of Notre Dame, Notre Dame, IN, USA

Fabiano Dalpiaz Department of Information and Computing Sciences, Utrecht University, Utrecht, The Netherlands

Eduard Paul Enoiu Mälardalen University, Västerås, Sweden

Jannik Fischbach Netlight Consulting GmbH, Munich, Germany; fortiss GmbH, Munich, Germany

Julian Frattini Blekinge Institute of Technology, Karlskrona, Sweden

Davide Fucci Blekinge Institute of Technology, Karlskrona, Sweden

Smita Ghaisas TCS Research, Pune, India

Jin L.C. Guo McGill University, Montreal, QC, Canada

Erik Kamsties Fachhochschule Dortmund, University of Applied Sciences and Arts, Dortmund, Germany

Clara Lüders Department of Informatics, University of Hamburg, Hamburg, Germany

Walid Maalej Department of Informatics, University of Hamburg, Hamburg, Germany

Daniel Mendez Blekinge Institute of Technology, Karlskrona, Sweden; fortiss GmbH, Munich, Germany

Lloyd Montgomery Department of Informatics, University of Hamburg, Hamburg, Germany

Nan Niu University of Cincinnati, Cincinnati, OH, USA

Fabian Panse Hasso-Plattner-Institut für Digital Engineering gGmbH, Potsdam, Germany

Cristina Ribeiro School of Applied Computing, Sheridan College, Oakville, ON, Canada

Mehrdad Saadatmand RISE Research Institutes of Sweden, Västerås, Sweden

Mehrdad Sabetzadeh University of Ottawa, Ottawa, ON, Canada

Anmol Singhal TCS Research, Delhi, India

Jan-Philipp Steghöfer XITASO GmbH IT & Software Solutions, Augsburg, Germany

Daniel Sundmark Mälardalen University, Västerås, Sweden

Sri Fatimah Tjong School of Computer Science, University of Nottingham Malaysia Campus, Semenyih, Selangor Darul Ehsan, Malaysia

Michael Unterkalmsteiner Blekinge Institute of Technology, Karlskrona, Sweden

Andreas Vogelsang University of Cologne, Köln, Germany

Jialiang Wei EuroMov Digital Health in Motion, Univ Montpellier, IMT Mines Alès, Alès, France

Jianzhang Zhang Hangzhou Normal University, Hangzhou Zhejiang, P.R. China

Liping Zhao University of Manchester, Manchester, UK

1. Handbook on Natural Language Processing for Requirements Engineering: Overview

Alessio Ferrari and Gouri Ginde

Abstract. Natural language processing (NLP) has played an important role in several computer science areas, and requirements engineering (RE) is not an exception. For more than 20 years, several works have been published on the application of NLP techniques to address RE-specific tasks, such as traceability, classification, defect detection and more. In recent years, the advent of massive and heterogeneous natural language (NL) RE-relevant sources, like tweets and app reviews, has increased the interest of the RE community in NLP. Furthermore, we witness a novel golden age of NLP technologies, enabled first by transformers, such as BERT, and more recently by large language models (LLMs), such as the GPT series and the Llama family. These developments offer the opportunity to solve long-standing RE tasks. Moreover, industrial case studies on NLP applications to RE problems also show that available NLP technologies are becoming increasingly industry ready. This book targets researchers and practitioners with a background in software engineering or information systems and presents a complete overview of the settled knowledge on NLP for RE. Its objective is to serve as a reference handbook for anyone interested in starting a journey in this field. This chapter introduces the book and provides an overview of its structure to guide the reader through the chapters.

Keywords: Natural language processing · Requirements engineering · NLP4RE · Large language models

A. Ferrari (✉)
University College Dublin (UCD), Dublin, Ireland

Consiglio Nazionale delle Ricerche (CNR), Istituto di Scienza e Tecnologie dell'Informazione "A. Faedo" (ISTI), Pisa, Italy
alessio.ferrari@ucd.ie, alessio.ferrari@isti.cnr.it

G. Ginde
Electrical and Software Engineering, Schulich School of Engineering, University of Calgary,
Calgary, AB, Canada
gouri.deshpande@ucalgary.ca

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1 Introduction

This book aims to provide a comprehensive overview of the research area of Natural Language Processing (NLP) for Requirements Engineering (RE), with the final objective of disseminating settled knowledge in the field and bringing new researchers and practitioners on board. Before starting the journey, it is good to provide a brief outline of what we mean by RE, what NLP is, what the scope of NLP for RE is and, ultimately, why and how you should read the rest of this book.

Requirements Engineering[1] is the branch of software engineering that focuses on the systematic process of discovering, documenting, analysing, verifying, validating and managing system and software requirements throughout the entire system life cycle. RE activities are traditionally dominated by Natural Language (NL), since this is the most widely understood communication code and easily fosters the exchange of knowledge among the different stakeholders who are involved in system development, from customers to developers and from business analysts to assessors.

What is a requirement? A requirement is a need that a system to be developed must satisfy for the benefit of at least one of the system's stakeholders. In a rigorous software development context, all of a system's requirements are typically documented in a Software or System Requirements Specification (SRS).

What is a software or system requirements specification? An SRS is a prescriptive document that defines what is expected from a system to be developed. In an agile development context, it is more common to express requirements in the form of user stories, which are reported in product backlogs. User feedback and issue tracking systems, as well as legal documents, can also be sources of requirements. Although requirements can be represented through visual and formal models, NL is still the most common way to write requirements in the industry, and this is not expected to change anytime soon.

What is the requirements process? The requirements process consists of several activities, which are typically iterative and often intertwined with software development. The initial one is requirements elicitation, in which requirements are identified from system-relevant stakeholders, from the examination of existing systems and documents or from an analysis of the market, e.g. through a comparison of existing products. This is followed by requirements analysis, which is a broad term encompassing various possible manipulations of the requirements, e.g. classifying, prioritizing and modelling, which bring towards the definition of an SRS or requirements document in general. Verification of an SRS's requirements ensures that the requirements are well formed. Validation of

[1] In this overview chapter, the absence of citations for the definition of this term and others is intentional. Section 2.4 provides citations to documents that define the main terms, such as Requirements Engineering and Natural Language Processing. For more specialized terms, e.g. POS tagging and issue tracking system, we provide definitions in two glossaries, one for NLP terms and one for RE terms; the glossaries are in Appendix A and B, respectively.

an SRS's requirements ensures that the requirements align with the actual needs and expectations of the stakeholders. Verification and Validation (V&V) techniques can include reviews or prototyping, to ensure that all requirements of an SRS are complete, consistent, realistic and unambiguous. The requirements management activity entails other tasks, such as guaranteeing traceability within an SRS and between an SRS and other artefacts, reusing requirements or processing user feedback to devise novel requirements and guide product improvement. These activities can be time consuming and error prone. Automated machine support can provide useful help to requirements and business analysts in performing them faster and with fewer errors.

Natural language processing is a multidisciplinary field at the intersection of artificial intelligence and linguistics. It focuses on enabling computers to process, understand, interpret and generate human language in a way that is both meaningful and useful. NLP encompasses a wide range of tasks that involve various aspects of language, including syntax (structure), semantics (meaning), pragmatics (context) and phonetics (sounds).

What are the typical NLP tasks? The typical NLP tasks are broadly categorized into two main types: those oriented towards text analysis and those oriented towards text generation. Among other tasks, text analysis includes text classification, which consists of assigning categories to a textual item, e.g. sentence or paragraph, as done in spam detection, sentiment analysis or topic categorization, and information extraction, such as Named Entity Recognition (NER), i.e. identifying proper nouns, such as names of people, organizations and locations, within a text. Text generation includes machine translation, i.e. translating a text from one language to another as in Google Translate or DeepL; dialogue generation, i.e. interacting with a user in NL, as typical in current chat-based Large Language Models (LLMs); and text summarization, i.e. creating a concise summary of a text.

What are the main NLP techniques? Several techniques and approaches are employed in NLP to achieve the aforementioned tasks. These include pre-processing techniques that prepare raw text data to be analysed and processing techniques that support the specific task at hand. Pre-processing techniques that prepare data are, among others, tokenization, i.e. splitting text into words or sentences; stemming and lemmatization, i.e. reducing words to their root forms; and vectorization, i.e. converting text into numerical representations, typically oriented to be processed with machine learning models. Processing techniques that support specific tasks can take various forms and can be composed of different processing elements that depend on the task at hand. NLP techniques can use rule-based or machine learning approaches. The former apply heuristic rules, e.g. "Whenever the term *happy* is encountered, classify the text as positive", in a sentiment analysis task. The latter use statistical models, typically based on machine learning strategies, that learn their expected behaviour from existing data given as examples.

Natural language processing for requirements engineering (NLP4RE) is the discipline that studies the application and evaluation of NLP techniques to solve RE tasks, analyse different requirements formats and types and process requirements-relevant artefacts. Let us now break down this definition.

What are NLP for RE tasks? Typical NLP for RE tasks include requirements classification; tracing and relating requirements with other requirements or software engineering artefacts, e.g. models, tests and code; requirements retrieval from large repositories; defect, i.e. smell or detection; and others.

Which requirements formats are available? Requirements can take different forms: "shall"-type requirements, e.g. "When the train driver presses the red button, the system shall disable the automatic train control system"; user stories, e.g. "As a train driver, I want to be able to enable door opening, so that passengers can drop off"; use case specifications, i.e. a system's behaviour as it responds to a sequence of actions from an external actor, outlining the interactions between the actor and the system to achieve a specific goal; and unstructured narratives.[2]

What are the different types of requirements? Requirements can have different types, also referred to as categories or classes. In particular, besides the classical distinction between functional *vs* non-functional requirements or qualities, several types of non-functional requirements exist. These include qualities such as performance, security, safety, availability and all the other attributes that a system is expected to exhibit—cf. the IEEE/ISO/IEC 29148-2018 International Standard [2] for a complete classification. In the NLP for RE literature, particular attention has been given to legal and privacy requirements.

What are requirements-relevant artefacts? Requirements-relevant artefacts are those from which one can extract requirements for a system, such as customer feedback provided in app reviews or social media networks or issues stored in issue-tracking platforms. Requirements-relevant artefacts also encompass regulations and norms—e.g. privacy policies and international product or process standards—as these include prescriptions that constrain system development and are thus relevant to requirements.

How to perform NLP for RE in practice? An RE task is typically complex to perform, while requirements documents and artefacts are hard to process with general-purpose NLP tools, given their domain specificity and project-dependent jargon. Due to these peculiarities, the design of sophisticated NLP pipelines is often needed in the NLP for RE field. NLP for RE tools also demands specific evaluation criteria that go beyond traditional NLP ones, as these are deemed insufficient to address the unique characteristics of the RE context. In addition, data management solutions are needed that take into account the intricacies of industrial requirements data, especially when these are processed by AI systems.

Why should I read this book? This book covers all these aspects and more, aiming to give a comprehensive overview of the settled knowledge in the NLP for RE field. We have asked many prominent researchers in the area to write chapters on their specific topics of research, oriented to introduce the reader to cutting-edge methods and findings. Each chapter delves into the nuances of a particular aspect of this interdisciplinary field, providing insights into the

[2]This book does not cover the last two types of requirements formats, as they have seen limited contributions in the field of NLP for RE.

theoretical foundations as well as practical applications. By bringing together the collective wisdom of these experts, the book aims to serve as a valuable resource for PhD students and researchers, equipping them with the knowledge and skills necessary to advance the state of the art in NLP for RE.

How should I read this book? First and foremost, you should read the current chapter to have an understanding of what you can expect from the content of the book. Then, you can jump to any of the other chapters. Although the chapters progress from fundamental RE tasks to more specialized and advanced topics, each chapter is designed to be an independent contribution, which can be read independently of the others. In addition, when a chapter uses a term that is unfamiliar to the reader, we provide Appendix A and B, which include the glossaries for the main NLP and RE terms, respectively. Readers who are not familiar with NLP or RE principles or who need to refresh their knowledge of some concepts are recommended to read these appendices before starting to read the rest of the book.

The rest of this chapter provides guidance for the reader to navigate and understand the book. Section 2 defines the book's objectives, its target audience, prerequisites and related publications. Section 3 explains the book's structure, with a brief outline of the book's chapters. Section 4 reports a brief biography of the editors.

2 Overview of the Book

2.1 Aims and Scope

The book aims to be a comprehensive handbook for the application of NLP technologies to RE. It first provides a detailed account of the different RE tasks that can be improved through the usage of NLP and describes the techniques that have been experimented with in RE. The book also covers different requirements types and artefacts and describes specific NLP techniques for their analysis. It finally gives guidelines on how to evaluate tools and perform NLP for RE in practice. As an appendix to the book, we provide an essential NLP glossary for readers to refer to when they encounter unfamiliar concepts. The final goal is to equip researchers and practitioners with a thorough understanding of how NLP can be effectively integrated into the RE process.

2.2 Target Groups

The book targets PhD students in software engineering and information systems, as well as researchers and practitioners, who want to have a comprehensive introduction to the RE tasks that can be automated using NLP and how to automate them. PhD students can benefit from a clear and comprehensive guide to the topic of NLP for RE. Researchers can identify further triggers for scientific exploration, based on the report of the currently settled knowledge on the topics covered by the book. This book can also serve as a valuable reference for researchers in NLP seeking to apply their techniques to RE. NLP researchers will benefit from an overview of RE tasks that heavily involve NL, which can be enhanced with their research. Additionally, practitioners grappling with challenges related to NL requirements can discover practical insights and essential background to enhance their RE processes with the assistance of NLP.

2.3 Prerequisites

Readers are expected to have a basic knowledge of software engineering, programming principles and RE. No specific NLP background is required since the book is specifically designed to be understood by subjects without prior experience in this field, and each chapter will introduce the necessary NLP concepts. However, it can be helpful to know the basic NLP principles and terminology. Appendix A, which provides an essential NLP glossary, can help in this sense.

2.4 Related Publications

No specific book publication is available on NLP for RE, and this makes the proposed book the first one on this topic. To get an overview of the RE field, a good starting point is the classical textbooks for RE, such as Klaus Pohl's *Requirements Engineering: Fundamentals, Principles, and Techniques* [5] or *Mastering the Requirements Process* by Robertson and Robertson [6], as well as practitioners-oriented RE books such as Dick et al.'s *Requirements Engineering* [1]. Other RE textbooks, such as van Lamsweerde's *Requirements Engineering: From System Goals to UML Models to Software Specifications* [7] or *Social Modeling for Requirements Engineering* [8], edited by Yu et al., are focused on specific RE model-based techniques, namely, KAOS and i*, and can be useful to delve into this topic. Another excellent reference is the recent online book *Requirements Engineering for Software Systems: A Concise Guide to Modern Ideas and Practices* by Letier.[3] Concerning NLP books, the main recommended readings are the fundamental references *Foundations of Natural Language Processing* [4], by Manning and Schütze, and *Speech and Language Processing* [3], by Jurafsky and Martin, which includes updated content on transformers and prompting.

3 Summary of the Chapters

The book is structured into 3 parts and 17 chapters. After the overview of the book—i.e. the current chapter—the first part, titled *NLP for Downstream RE Tasks*, illustrates the different RE tasks to which NLP can be applied, such as classification, traceability, defect detection, etc. The main focus of this part is on typical "shall" requirements, as these are mostly used in domains that follow a requirements-centric process, e.g. transportation or healthcare, in which several RE tasks are needed to ensure requirements quality. Furthermore, "shall" requirements have historically been the focus of automated support in the literature.

[3] http://www0.cs.ucl.ac.uk/staff/E.Letier/book/. Accessed July 10th, 2024.

The second part, *NLP for Specialized Types of Requirements and Artefacts*, focuses on approaches for the analysis of different requirement types, such as legal and privacy requirements; requirements formats, with a particular focus on user stories; and requirements-relevant artefacts, e.g. app reviews and information from issue tracking systems. The third part, *NLP for RE in Practice*, provides an overview of NLP tools specialized for RE tasks, as well as guidelines to support the evaluation of NLP for RE tools, the usage of LLMs and the mitigation of data-centric challenges emerging from the application of NLP to RE problems. In the following, we provide a summary of the different chapters.

Chapter 1—Handbook on Natural Language Processing for Requirements Engineering: Overview The chapter provides an overview of the book. It first gives high-level definitions for RE and NLP and explains the fundamental concepts required to understand the rest of the volume. It then outlines its scope, target audience, prerequisites and related publications. Finally, it guides the readers throughout the subsequent chapters, summarizing their content and explaining the role of the explained task, artefacts and techniques in the RE process.

Part I—NLP for Downstream RE Tasks

Chapter 2—Machine Learning for Requirements Classification When large-scale products are developed, also large requirements sets are produced, which need to be organized and classified. Classifying requirements implies associating each requirement with a set of labels that give semantic information about the requirement. The labels can specify the type of requirement (functional or non-functional), its functional category (communication protocol, user interface, etc.) and its non-functional category (security, availability, maintainability, usability, etc.). Classification helps handle large amounts of requirements and drives the apportionment of requirements to specific software components. NLP techniques for text classification can be applied to speed up requirements labelling and organization. This chapter provides an in-depth overview of NLP and machine learning techniques in general to perform requirements classification. It defines the problem and introduces several fundamental concepts, including text representation, evaluation metrics and machine learning algorithms, which will also be used in the rest of the book. Finally, it provides an overview of past research in automated requirements classification using different machine learning paradigms.

Chapter 3—Requirements Similarity and Retrieval Companies typically specialize in the production of systems for a certain set of domains, e.g. aerospace or healthcare. When developing novel products, it is a good and economically beneficial practice to reuse requirements and code. This also occurs when a company aims to reuse existing requirements and software process artefacts to align with the specifications of public procurement documents, as is typical in tenders for domains such as public transportation, energy, telecommunications and others. In all these cases, automatically retrieving requirements and assets from company repositories according to novel product features or requirements can improve the reuse practice. NLP-based information retrieval approaches, typically leveraging the computation of the linguistic similarity between requirements or other artefacts, are the natural solutions for these types of situations. This chapter introduces the concept of linguistic similarity and several NLP-based techniques for data representation and similarity computation. Finally, the chapter demonstrates the practical application of these techniques in requirements retrieval for the reuse of requirements and code.

Chapter 4—Natural Language Processing for Requirements Traceability Along the software process, requirements are justified, re-negotiated, discussed, refined and gradually evolved into executable artefacts. The trace that this evolution leaves has to be identified and controlled, to ensure that each high-level requirement agreed with the customer is implemented in the product. Traceability is the discipline of linking requirements artefacts to other requirements, possibly at a different level of abstraction, and to other artefacts—models, software components and tests—of the software process. NLP can support automated identification of the trace links, also by suggesting possible links that the analyst may have neglected. This chapter introduces the field of requirements traceability, describes four main tasks in this area—namely, trace link recovery, maintenance, explanation and link type prediction—and discusses how to address them through NLP techniques. The chapter also considers current limitations and future directions for research, especially associated with the usage and evaluation of generative AI approaches.

Chapter 5—Detecting Defects in Natural Language Requirements Specifications Requirements specifications are the cornerstone of communication within the software process, and they are expected to be unequivocal enough to be interpreted in the same way by stakeholders who deal with them and to drive towards a consistent implementation of the system. Therefore, NL defects in the requirements, such as ambiguity and inconsistency, have to be detected by means of time-consuming reviews. These can be made more effective by means of NLP techniques, which allow analysts to easily discard defects and improve requirements quality. This chapter highlights the types of defects that can arise in NL requirements, with a particular focus on ambiguity, and explains how defect-detection tools function. The chapter also reviews the empirical evaluation of defect-detection tools, focusing on two main aspects: effectiveness, measured by how well the tool detects defects within its scope, and usefulness, assessed by whether detecting these defects improves the quality of the resulting system. It observes that while many defect-detection tools are accompanied by demonstrations of their effectiveness, evidence of their usefulness is often lacking. The chapter concludes with a reflection on the utility of defect-detection tools, provocatively questioning whether they genuinely enhance the overall quality of the system to be developed.

Chapter 6—Automated Requirements Terminology Extraction The text of requirements is typically highly specialized in terms of jargon that it uses, which is frequently domain-dependent and even project-dependent, with acronyms and multi-word terms that are familiar only to expert software engineers. Identifying these terms from requirements can help create glossaries that guide the understanding of requirements specifications and create a common understanding between stakeholders involved in the RE process. Furthermore, extracting key terms can also help generate models from the requirements. Support from NLP techniques can alleviate the time-consuming and error-prone work of manually extracting requirements terminology. The chapter introduces relevant definitions and typical application scenarios of terminology extraction and then describes the different families of techniques to address this task, including machine learning, rule-based and statistical methods, together with current challenges and limitations.

Chapter 7—Automated Requirements Relations Extraction In a requirements document, requirements are typically related to one another. For example, the fulfilment of one requirement may depend on the satisfaction of another requirement. A low-level requirement may refine a high-level one to provide better guidance for implementation. Additionally, multiple requirements can contain similar or duplicate content, indicating redundancy. Correctly identifying these relations, possibly with the support of NLP solutions, can provide better control over the RE process, support traceability and ensure that constraints between requirements are maintained during requirements evolution. The chapter provides a comprehensive overview of the topic, starting with the fundamentals of requirements relations and the most common types. It is then structured around two main sections: the first covers NL techniques for the representation of requirements (syntactic vs. semantic techniques), while the second delves into information extraction methods for relation extraction (retrieval-based vs. machine learning-based methods). The analysis is complemented with an example application of the techniques and a discussion on challenges and future research directions.

Part II—NLP for Specialized Types of Requirements and Artefacts

Chapter 8—Legal Requirements Analysis: A Regulatory Compliance Perspective Numerous types of software and applications need to comply with specific regulations due to the sensitive nature of the data they handle or the critical functions they perform. These include systems developed in different domains, such as healthcare, finance, eCommerce, transportation and others. Regulations typically include legal requirements that the system is expected to satisfy. NLP technologies can help also in this case, by extracting software-related rights and obligations from existing regulations, thus reducing the human effort to identify requirements from complex legal documents. NLP can also help ensure that specifications are compliant with legal norms, thereby supporting the work of certification authorities and developers. The chapter explores methods for analysing legal requirements, with GDPR serving as an exemplar. It first presents how to abstract and capture legal knowledge through suitable machine-analysable representations based on conceptual models. Then, it presents an end-to-end NLP pipeline for information extraction from legal documents and compliance verification. The chapter is complemented by a reflection on challenges in legal compliance checking and future research directions.

Chapter 9—Privacy Requirements Acquisition and Analysis Privacy requirements are a particular case of legal requirements. These are especially relevant in today's context, in which Web and mobile applications leverage users' personal data to provide services and better target advertisements. When developing software for public use, developers need to identify and implement these types of requirements. Privacy requirements can be included in different types of documents, such as regulations, standards and privacy notices, and NLP solutions for information extraction can be used to automate their identification and facilitate developers in the development of privacy-compliant applications. This chapter reviews different techniques for extracting privacy requirements from texts and demonstrates how these requirements can be utilized for the analysis of both requirements and software. It also outlines the challenges associated with ambiguity, vagueness, generalization and incompleteness in NL, underscoring their significance in compliance analysis. The chapter concludes by presenting three downstream tasks exemplifying how extracted privacy requirements can be employed to scrutinize software systems, encompassing data flow tracing, source code compliance checks and consent verification within evolving NL policies.

Chapter 10—On the Automated Processing of User Feedback Once the product is developed, users may express their feedback on the product, and the feedback may include additional requirements to be implemented to better satisfy users' needs. Mobile apps receive large amounts of feedback in the form of reviews, and NLP techniques can be used for several purposes, e.g. to distinguish reviews that conceal requirements from generic comments and bug reports and identify features that need to be enhanced based on users' expressed concerns. Other sources of feedback are social media platforms and product forums, which also help gain insights into user behaviour and opinions. However, leveraging the full potential of user feedback poses challenges, primarily related to the large quantity of data and varying quality of feedback. The chapter presents the task of user feedback analysis and how this can be useful to understand user needs, monitor and improve product quality, plan project development and help users, e.g. through chatbots. It then describes an NLP pipeline for processing user feedback, based on text classification, clustering and summarization. Finally, the chapter outlines various dimensions of the data quality management problem and provides concluding remarks, emphasizing the need to develop additional approaches to address the quality issue.

Chapter 11—Mining Issue Trackers: Concepts and Techniques During the development of a software product, several issues may arise that need to be reported and managed to ensure that the process is under control. In today's development, all these issues are frequently stored in so-called issue tracking systems, which are platforms that enable internal and external stakeholders to collaboratively report and discuss bugs, novel features, user-reported problems and other information, including requirements. Using NLP to navigate these complex and data-intensive systems can be useful for various purposes, such as analysis of issue quality, monitoring of product evolution and ultimately providing an informed view of the status of system development. This chapter introduces the concept of issue trackers, how their analysis can be useful in software development and what are the fundamental characteristics of these platforms. It then illustrates different use cases of mining issue trackers—including issue quality analysis, evolution and discussion analysis and structure analysis—together with NLP techniques to address them. Each use case is complemented with pitfalls and takeaways that highlight best practices for applying existing techniques in real-world contexts.

Chapter 12—Automated Analysis of User Story Requirements Agile processes have become one of the most common software development practices. In these contexts, it is common to express requirements in the form of so-called user stories, a NL format that synthetically expresses desired user needs and associated rationales. These can be particularly numerous, and it can be hard to have a holistic view of the expected system behaviour, which is particularly needed in a rapidly changing context such as the agile one. NLP techniques for information extraction can help provide high-level and domain models based on user stories, give developers and stakeholders the comprehensive understanding they need and form the basis for information exchange. This chapter focuses on NLP techniques for user story analysis. A quality framework is presented to classify defects in the formulation of user stories, together with an NLP-based tool called AQUSA for defect identification. Additionally, the Visual Narrator tool is introduced, which extracts a domain model from a collection of user stories using heuristics assembled from the literature. The chapter concludes by outlining ongoing and future directions for the use of NLP techniques in conjunction with user stories and related artefacts.